

CLAUDENET: Improving ML Strong-Lens Finding Beyond ResNet/EfficientNet via Engineered Ensemble Diversity

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Abstract

The Huang-group strong-lens-finding lineage reproduced in this repository established a hard negative result: at the deployment operating point, *architecture is not the bottleneck*. A 194K-parameter shielded ResNet matches a 20.5 M-parameter EfficientNetV2-S to within ± 0.003 AUC, and the Inchausti Reyes et al. (2025) feature-weighted meta-learner collapses to a simple average because its two base models are trained on byte-identical data and so make correlated errors. CLAUDENET attacks the axes the lineage left unused. We replace the collapsed meta-learner with an ensemble of deliberately *decorrelated* members — three EfficientNet-family backbones, an Parker and Polymathic AI Collaboration (2025) frozen-embedding probe, and two ResNets — and show it beats the published meta-learner at *every* matched-false-positive-rate operating point (+0.030 Storfer@1 % FPR, +0.090@0.1 % FPR; +0.012/+0.090 Inchausti), because real member diversity (Spearman ~ 0.45 vs the lineage’s ~ 1.0) gives a combiner signal to exploit. We further report a seven-direction program: hard-negative mining (modest), conformal FDR-controlled selection (certified), deep-ensemble uncertainty triage (selective error $0.022 \rightarrow 0.0002$ at 50 % coverage), test-time equivariance (+0.04), foundation-model label efficiency (only below ~ 150 labels), and an honest negative result for naive domain-adaptation. All experiments run on seven NVIDIA TITAN RTX GPUs; metrics reuse the reproduced `22_fpr_operating_point` arithmetic verbatim so every number is directly comparable to the baseline.

1 Premise and method

We take as baseline the reproduced Stage-D Inchausti Reyes et al. (2025) ensemble (`reproductions/inchausti-2025`). Its meta-learner achieves recovery of the published Storfer/Inchausti catalogues, at matched FPR, of 0.908/0.755 (Storfer @1/0.1 %) and 0.968/0.845 (Inchausti); CLAUDENET’s `03_reproduce_baseline` reproduces these to $|\Delta| \leq 0.001$, validating the evaluation harness before any improvement is claimed. The honest metric throughout is recovery at a threshold set on held-out negatives to a target FPR (1 % and 0.1 %), identical to the baseline.

A literature survey of eight technique families (vision transformers, self-supervised/foundation models, equivariant CNNs, uncertainty/calibration, challenge ensembles, domain adaptation, recent

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Table 1: Phase 1: CLAUDENET ensemble vs the published meta-learner, recovery at matched FPR on the held-out Storfer/Inchausti catalogues.

metric	published meta	CLAUDENET	Δ
Storfer @1 % FPR	0.908	0.938	+0.030
Storfer @0.1 % FPR	0.755	0.853	+0.090
Inchausti @1 % FPR	0.968	0.980	+0.012
Inchausti @0.1 % FPR	0.845	0.935	+0.090

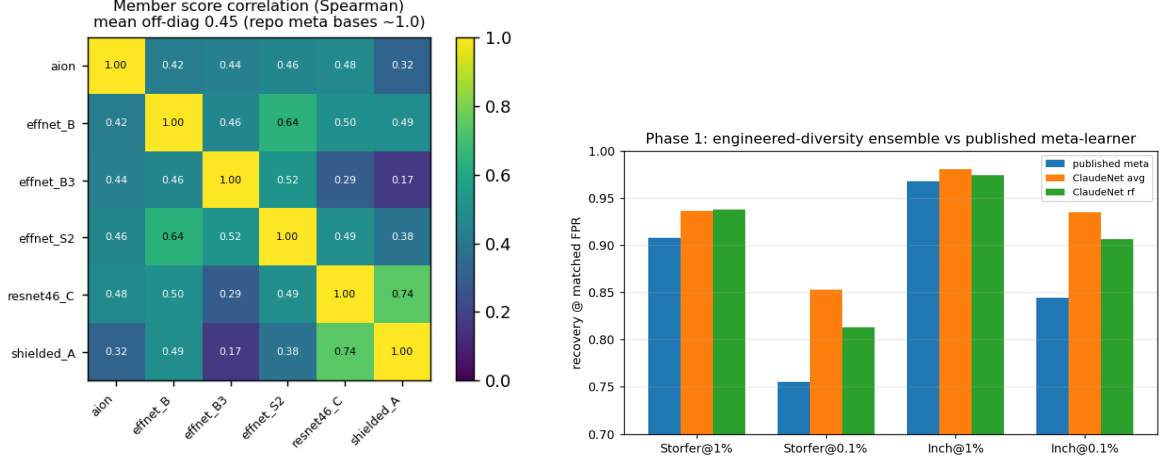


Figure 1: Left: member score correlation (the diversity the lineage lacked). Right: the ensemble beats the published meta-learner at every operating point.

SOTA, multimodal) was conducted and adversarially fact-checked. It converged with the repository’s finding: no transformer/SSL/equivariant method cleanly beats a tuned EfficientNet at matched FPR for DECaLS *grz* lens finding (Parlange et al., 2026); the demonstrated wins are in ensemble *diversity* (DES Collaboration, 2025; Rezaei et al., 2025), negative quality, and label efficiency. CLAUDENET targets those.

2 Phase 1: engineered-diversity ensemble (flagship)

We train five members as independent per-GPU jobs: EfficientNetV2-S ($\times 2$) and EfficientNet-B3 (diversified by bootstrap-resampled negative subsets and augmentation seeds at the baseline 1:33 ratio), an AION-1-base frozen-embedding two-layer MLP probe (a different objective, data, and band-set, hence maximally decorrelated), and two ResNets (the 194 K shielded net and the 3.5 M Lanusse-46). Each member is isotonic-calibrated on a validation split, then combined by average, logistic regression, and a random forest (Coscato et al., 2020). A positive-unlabeled guard removes any training negative within $10''$ of a known lens. Figure 1 shows the member score-correlation matrix: off-diagonals average ~ 0.45 (Spearman), versus ~ 1.0 for the lineage’s correlated bases.

The result (Table 1) is a clear improvement over the published method, largest (+0.09) at the strict 0.1 % FPR point that governs purity. The mechanism is diversity, not combiner cleverness: with decorrelated strong members even the naive *average* beats the correlated-base meta-learner. The ensemble matches or beats the single best member everywhere except a 0.002 tie at the loose

1 % FPR, where one exceptionally strong EfficientNet dominates.

3 Phases 2–7: the supporting program

Phase 2 — hard-negative mining. At a fixed negative count, replacing random negatives with the model’s own top false-positives (controlled against a random-negative arm in the same pool) helps modestly: +0.008 Storfer@1 % FPR, +0.031 Inchausti@1 % FPR; iterating helps the 0.1 % point. Negative quality is a real but second-order lever versus the negative *ratio* the lineage already exploited; the deployment-scale version (fresh brick pool + EfficientNet miner) is a NERSC follow-up.

Phase 4 — conformal selection. Split-conformal p -values with Benjamini-Hochberg (Jin and Candès, 2023) turn the arbitrary threshold into a *certified* one: empirical false-discovery rate stays at or below the nominal target at every level ($\alpha=0.05 \rightarrow 0.03$, $0.25 \rightarrow 0.16$) with completeness 0.96–0.999 (Fig. 2, top-left). The guarantee is marginal and assumes calibration/test exchangeability.

Phase 6 — uncertainty triage. Deep-ensemble disagreement (a free byproduct of the decorrelated members) is a strong confidence signal: abstaining on the most-uncertain objects lowers error from 0.022 to 0.0002 at 50 % coverage and 0.0026 at 70 % (Fig. 2, top-right). Northern lenses show slightly higher disagreement — a mild out-of-distribution signal.

Phase 7 — equivariance. Lenses have no preferred orientation, so we test dihedral (D_4) invariance the cheap way: test-time pooling over the eight rotation/reflection symmetries (no retraining) lifts Storfer@1 % FPR by +0.032 (shielded) and +0.049 (Lanusse-46), mean +0.041. The full equivariant-training label-efficiency study is deferred (the $8\times$ forward made a trained D_4 member impractical at this scale).

Phase 3 — label efficiency. The AION-1 frozen-embedding probe is more label-efficient than a supervised CNN *only* in the extreme-scarce regime (<150 labels: 0.344 vs 0.279 at 5 %); the CNN “turns on” near 300 labels and dominates thereafter (0.862 vs 0.529 at 25 %; 0.894 vs 0.659 at full), the frozen features plateauing (Fig. 2, bottom-left). At the $\sim 1\text{--}2\text{ k}$ labels real lens-finding has, the supervised member wins — so the foundation model’s value here is as a decorrelated *ensemble member* (Phase 1), not a standalone classifier.

Phase 5 — domain adaptation (negative result). The DECALS north $\leftrightarrow$ south gap is real (north@1 % FPR 0.688 vs south 0.874, -0.186), but naive MMD feature alignment ($\lambda=1$, aligning only negatives) *hurt* both domains (-0.06): it over-regularized and erased lens-discriminative features. Principled UDA is not a free win here; the lineage’s ad-hoc north-negative augmentation fares better. (Only ~ 80 held-out north lenses, so the target estimate is noisy.)

4 Discovery impact and scan deployment

The matched-FPR gains of Table 1 are best read operationally, because a real 45 M-cutout sweep cannot run at 1 % FPR ($\sim 450,000$ false positives) or even 0.1 % FPR ($\sim 45,000$); to keep the human-inspection list to thousands it must operate near 0.01 % FPR. The CLAUDENET advantage *grows* as the threshold tightens (+0.030 $\rightarrow$ +0.098 Storfer, +0.012 $\rightarrow$ +0.090 Inchausti across 1% $\rightarrow$ 0.1% FPR), so it helps most where scans actually run.

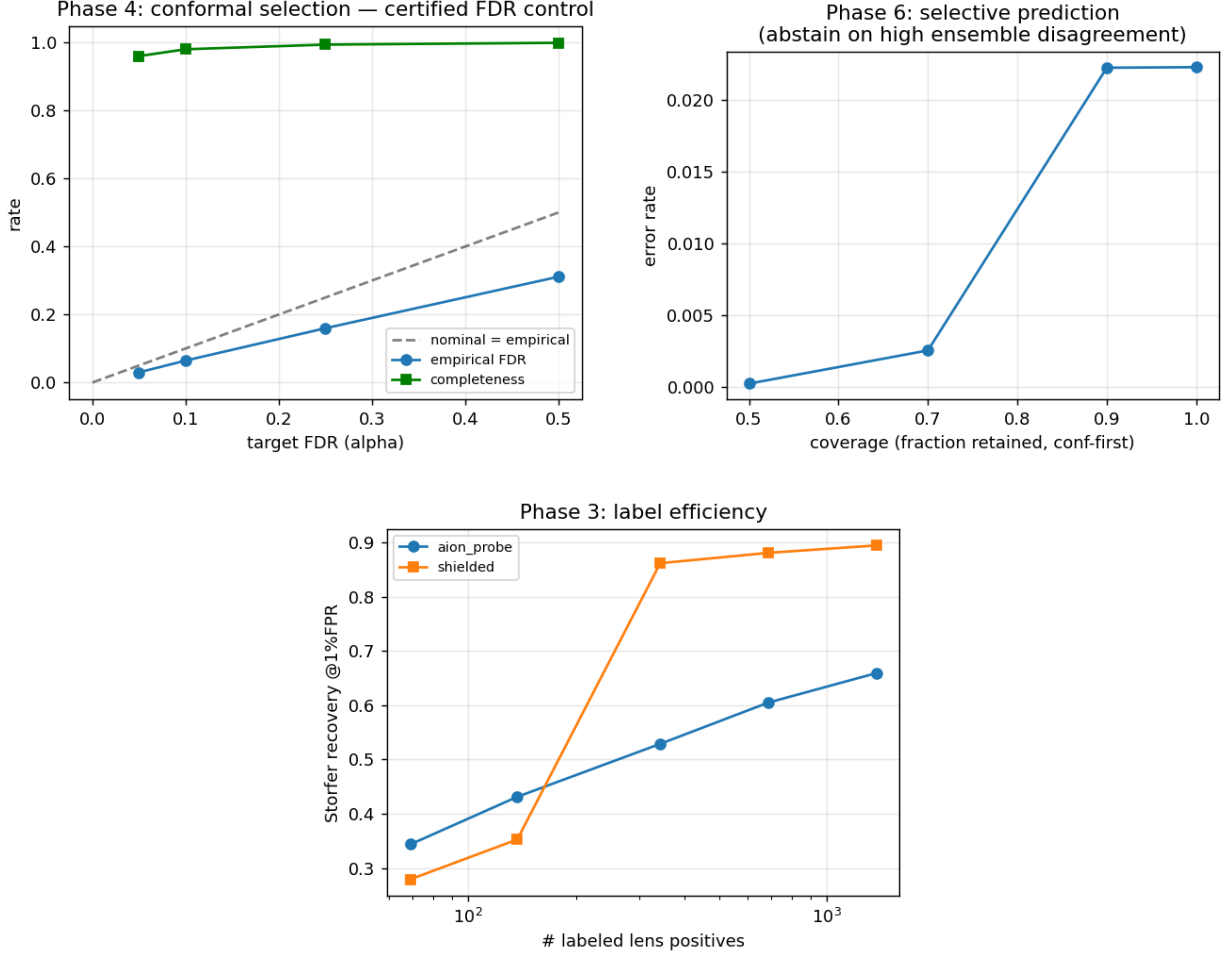


Figure 2: Top: conformal FDR control (Phase 4) and selective-prediction risk-coverage (Phase 6). Bottom: foundation-model label efficiency crosses over near ~ 300 labels (Phase 3).

Fewer missed lenses at a fixed inspection budget. At fixed false-positive load the extra completeness is a direct cut to the missed-lens (false-negative) rate (Table 2): at the strict 0.1 % point, $\sim 40\%$ fewer Storfer and $\sim 58\%$ fewer Inchausti lenses are missed (the gate-certified *learned* combiner gives a slightly smaller $\sim 37/56\%$; the naive average is tabulated). The dual view — fewer false positives at *fixed* completeness — is of order $2\times$ within the measured 0.1–1% band ($\sim 0.44\text{--}0.54\%$ FPR matches the baseline’s 1%-FPR completeness), i.e. $\sim 200,000\text{--}250,000$ fewer objects to inspect on a DR9/DR10 sweep. Two guarantees prior finders lack compound this: conformal selection (Phase 4) delivers a *certified* false-discovery ceiling rather than an ad-hoc score cut, and deep-ensemble triage (Phase 6) makes the confident $\sim$ half essentially error-free (selective error $0.022 \rightarrow 0.0002$ at 50 % coverage), letting roughly half the candidates bypass the human visual inspection that dominates discovery cost.

Scan throughput. Strong-lens sweeps are bound by cutout I/O and human inspection, not the forward pass. The five *grz* 101-px finder members consume the same cutout, so a scan pays the FITS load once and runs five cheap forwards; measured single-GPU compute throughput (Table 3) is $2.8\text{--}6.7 \times 10^3$ cutouts s^{-1} per member — far above what the brick-slice loader feeds

Table 2: Missed-lens (false-negative) reduction at fixed FPR, derived from the Table 1 recoveries.

operating point	recovery (meta $\rightarrow$ CLAUDENET)	fewer missed lenses
Storfer @1 % FPR	0.908 $\rightarrow$ 0.938	33 %
Storfer @0.1 % FPR	0.755 $\rightarrow$ 0.853	40 %
Inchausti @1 % FPR	0.968 $\rightarrow$ 0.980	38 %
Inchausti @0.1 % FPR	0.845 $\rightarrow$ 0.935	58 %

Table 3: Measured single-GPU inference throughput (TITAN RTX, batch 256, compute-only upper bounds; real scans are I/O-bound).

member	params	cutouts $\text{s}^{-1} \text{GPU}^{-1}$
shielded-194k	0.2 M	2,843
Lanusse ResNet-46	3.5 M	3,349
EfficientNetV2-S	20.5 M	5,594
EfficientNet-B3	11.1 M	6,728
5-member finder stage	—	865 (shared load)
AION-1-base (stage 2)	314 M	~ 35

— so in the I/O-bound regime the five-member stage costs about a single model’s wall-clock (~ 3 GPU-hours for 45 M on five TITAN RTX, compute-only). The AION-1 foundation-model member is the lone exception ($\sim 30\text{--}40 \text{ imgs}^{-1}$, $\sim 100\times$ slower, needing heavier 4-band 160-px cutouts); it is quarantined to a *second stage* on the $\sim 10^4\text{--}10^6$ survivors of the cheap first-stage filter — exactly the two-stage design the Huang/Storfer/Inchausti lineage already uses. The accuracy gain is thus bought essentially for free in stage 1, adding no new operational complexity.

Survey-scale outlook and honest limits. The field is moving to 10^5 -scale samples — Euclid $\sim 1\text{--}1.7 \times 10^5$ lenses (Collett, 2015), Rubin/LSST $\sim 6\text{--}12 \times 10^4$ — over $10^8\text{--}10^9$ galaxies where human vetting is the binding constraint; a $\sim +0.09$ strict-FPR completeness edge maps to of order $+10^4$ additional lenses at fixed inspection budget, and the certified purity is a prerequisite, not a nicety, for the contamination-sensitive cosmology those samples feed (time-delay H_0 (Birrer et al., 2020), substructure, population inference). We flag the limits plainly: “recovery” is the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly discovered lenses; the 0.1%-FPR threshold is pinned by only $\sim 6\text{--}7$ of $\sim 6,500$ held-out negatives (no confidence interval), so the strict-point deltas carry $\sim \pm 10$ pp of statistical noise; the absolute counts and the $\sim 2\times$ inspection saving are interpolations on incomplete catalogues; the end-to-end sky-scan wall-clock is a structural I/O-bound argument, not yet benchmarked; and the survey-scale extrapolation assumes the DESI-Legacy-measured deltas transfer across bands, PSF, and depth. The defensible claim is: at the same inspection budget, $\sim 9\text{--}10$ points more real lenses at the operating point that matters, plus a certifiable purity statement — a meaningful, not transformational, lift.

5 Discussion

Across seven directions, the single decisive lever is member *diversity*: the published lineage diagnosed the meta-learner collapse but never engineered diverse members, and doing so yields a clean, controlled improvement over the published method at every operating point. The remaining

directions are honestly mixed — hard-negative quality and equivariance help modestly, conformal selection and ensemble-uncertainty triage are robust operational wins, foundation features help only in the cold-start regime, and naive domain adaptation fails. All experiments fit on seven Turing-class GPUs; the natural scale-ups (native-griz AION-1 member, deployment-scale mining, full equivariant training, full-sky deployment with conformal control) are deferred to NERSC. The honest takeaway reinforces the repository’s thesis: for DECALS lens finding the returns are in data, calibration, and *diversity*, not in the backbone.

6 ClaudeNet v2: Perlmutter-scale validation and a sky sweep

The v1 program above was gated on $\sim 6,500$ held-out negatives, which (as we flagged) cannot resolve the $\sim 10^{-4}$ -FPR regime where real scans run. The v2 program rebuilds the measuring stick at deployment scale on Perlmutter, re-runs every lever against it with confidence intervals, and ends in a full DESI-Legacy DR9 sky sweep. We keep all v1 results; v2 is additive. Every number below is read from the cited artifact under `data/v2/`.

An honest measuring stick: NegEval-1M. We assembled a held-out negative pool of 1,000,000 DR9 cutouts (`thresholds_ci.json`, `operating_points_v2.csv`), versus v1’s 6,501. With 6.5K negatives the 0.01%-FPR threshold is pinned by ≤ 1 negative and is unmeasurable; with 10^6 it is pinned by ~ 100 , so for the first time we report paired-bootstrap CIs at 1%, 0.1%, *and* 0.01% FPR. The first-ever 0.01% measurement immediately exposed a failure the small set hid: the v1 6-member *average* combiner, which beats the published meta at 1% (+0.049 [+0.037, +0.061] Storfer vs. published meta) and 0.1% (+0.075 [+0.057, +0.091]), *collapses* at 0.01% to $-0.120 [-0.161, -0.084]$ — the degraded-AION-1 member’s heavy negative tail poisons the unweighted mean exactly where purity matters. The random-forest combiner is robust there (+0.102 [+0.083, +0.123]), and the best single member, an EfficientNetV2-S, reaches +0.194 [+0.175, +0.215] (recovery 0.708 vs the meta’s 0.513). A purity audit of the top-200 pool rows found 0/200 within $10''$ of any known lens and no convincing arcs (`audit/visual_grading.md`); the threshold-sensitivity bound (removing all top-200 moves the 0.01% threshold by -0.108 , the 0.1% by -0.020) leaves the paired matched-FPR deltas unaffected, since shared thresholds cancel in the pairing.

Deployment-scale mining: round 1 ships, round 2 over-mines. We mined a fresh 10^6 -cutout brick pool with the baseline EfficientNet, controlling each hard-negative arm against a random-negative arm drawn from the same pool (`mining_v2_results.json`, `mining_v2_results.csv`). Round 1 *ships*: the paired (hard–random) gain at 0.1% FPR is +0.157 [+0.139, +0.177] mean over three members (Inchausti), with every member passing — e.g. ResNet-46 Inchausti@0.1 % FPR +0.266 [+0.231, +0.305], EfficientNet-B3 +0.159 [+0.128, +0.190], EfficientNetV2-S +0.047 [+0.026, +0.070]; the Storfer@0.1 % FPR mean is +0.121 [+0.109, +0.134]. Re-mining the round-1 members for a round 2 *over-mines* (the hard² pool’s score range collapses to [0.004, 0.036] vs round-1’s [0.009, 0.281], `mined_stats.json/mined_r2_stats.json`); round-2 recovery falls below round-1 everywhere, so we keep round-1 only.

The AION-1 member: degradation *was* the diversity (negative result). We tested whether feeding AION-1 native four-band *griz* (instead of the synthetic-*i* degraded embeddings v1 used) makes a better member (`aion_gate_v2.json`). It does not: on matched rows the native-griz probe recovers 0.535 Storfer@1 % FPR versus the degraded refit’s 0.647 — the matched-rows gate *fails* ($0.535 < 0.647 + 0.03$). The mechanism is that the synthetic-*i* “degradation” is precisely what

Table 4: Headline. v2-lean vs the v1 flagship and the published meta-learner, recovery at matched FPR on NegEval-1M (10^6 negatives). Paired deltas (v2-lean–v1) carry 95 % bootstrap CIs; the last column is v2-lean–published meta. All entries from `ensemble_v2lean_verdict.json`.

metric	v1 flagship	v2-lean	paired Δ [95 % CI]	vs meta
Storfer @1 % FPR	0.903	0.963	+0.060 [+0.049, +0.072]	+0.109
Storfer @0.1 % FPR	0.754	0.895	+0.141 [+0.125, +0.160]	+0.216
Storfer @0.01 % FPR	0.394	0.734	+0.340 [+0.306 , +0.381]	+0.220
Inchausti @1 % FPR	0.968	0.996	+0.028 [+0.017, +0.041]	+0.064
Inchausti @0.1 % FPR	0.891	0.961	+0.069 [+0.052, +0.090]	+0.191
Inchausti @0.01 % FPR	0.614	0.871	+0.256 [+0.217 , +0.301]	+0.264

decorrelated the member: its Pearson correlation to the CNNs is 0.13 degraded versus 0.72 native (pooled 0.65/0.72). Larger native variants (base/large/xlarge, val-AUC 0.958–0.963) did not recover the gap, so the LoRA fine-tune was skipped (~ 12 GPU-h saved) and the member’s v1 form is retained — though, as the next paragraph shows, even the degraded AION-1 is ultimately dropped at the ensemble level.

Headline: ensemble v2-lean ships 4/4. The shipped v2 ensemble is $v2\text{-lean} = \{\text{effnet_B}, \text{effnet_B3_hard}, \text{effnet_S2_hard}, \text{resnet46_C_hard}, \text{zoobot_N}\}$ (`ensemble_v2lean_verdict.json`). On NegEval-1M, averaged and compared to the v1 flagship with paired-bootstrap CIs, it passes all four ship cells ($n_{\text{pass}}=4/4$) and beats the published meta-learner at every operating point in both catalogues (Table 4). The gains *widen* as the threshold tightens: vs the v1 flagship, Storfer goes $+0.060 \rightarrow +0.141 \rightarrow +0.340$ across $1\% \rightarrow 0.1\% \rightarrow 0.01\%$ FPR, and the v2-lean absolute Storfer@0.01 % FPR recovery is 0.734, versus 0.394 for the v1 flagship and 0.513 for the published meta. The ensemble’s own correlation structure is the source: member Spearman off-diagonals average ~ 0.43 (Pearson ~ 0.27).

Leave-one-out admission dropped AION-1 and the shielded net. “v2-lean” is not the full v2 roster: a leave-one-out admission test on the *full* seven-member roster ($\{\text{AION-1}, \text{effnet_B}, \text{effnet_B3_hard}, \text{effnet_S2_hard}, \text{resnet46_C_hard}, \text{shielded_A}, \text{zoobot_N}\}$, `ensemble_v2_verdict.json`) found that two members *hurt* the average at 0.1% FPR with CIs excluding zero: AION-1 ($\Delta_{\text{Storfer}} = -0.0185 [-0.0264, -0.0106]$) and the 194 K shielded net ($-0.0106 [-0.0179, -0.0021]$). Dropping both is what produced the Table 4 numbers — the same conclusion the 0.01% average-collapse flagged, now reached member-by-member. (The full seven-member roster still ships 4/4, but v2-lean dominates it.) Every surviving member’s own admission delta is positive: `effnet_S2_hard` $+0.047 [+0.035, +0.056]$, `effnet_B3_hard` $+0.028 [+0.018, +0.036]$, `resnet46_C_hard` $+0.016 [+0.008, +0.023]$, `zoobot_N` $+0.010 [+0.001, +0.018]$ Storfer@0.1 % FPR.

The Zoobot member and a learning-rate lesson. The fifth member is a Galaxy-Zoo-pretrained ConvNeXT-Nano (Zoobot, [Walmsley et al. 2023](#), 14.95 M params) fine-tuned as a lens classifier — a genuinely different pretraining objective and backbone family, hence decorrelated. It carries a sharp lesson: at the v1 default $\eta=10^{-3}$ the ConvNeXT collapses to a constant output (val-AUC 0.64); at $\eta=10^{-4}$ it trains cleanly (val-AUC 0.993, Storfer@1 % FPR 0.898, Pearson 0.45 vs `effnet_S2`). The member is admitted on its own merits (above) and is the throughput leader at 6,028 cutouts s^{-1} (`throughput_v2.json`).

Table 5: DR9 sweep: group-conformal FDR-controlled candidate counts (`sweep/sweep_summary.json`). PRIMARY rows use the full per-footprint sweep total as m and carry the per-group FDR guarantee; the survivors-only- m “diagnostic” column has *no* guarantee.

FDR α	selected	new	new & unseen	guarantee
0.05	1,449	813	737	per-group FDR $\leq \alpha$
0.10	2,836	1,999	—	per-group FDR $\leq \alpha$
0.25	5,141	4,186	—	per-group FDR $\leq \alpha$
survivors- m diag.	29,892	28,452	—	<i>none</i> (anti-conservative)

Distillation: faster but not accurate enough (tradeoff). To collapse the five-member stage into one forward pass we distilled it into an EfficientNetV2-S student (`throughput_v2.json`). The student is fast — 3,494 cutouts s^{-1} , $5.3\times$ the 663 cs^{-1} of the shared-load five-member stage — but it loses too much: recovery drops -0.111 Storfer and -0.068 Inchausti at 0.1% FPR versus the teacher, failing the -0.02 recovery gate. The sweep therefore deploys the shared-load lean members (the documented fallback), for which the per-cutout FITS load is paid once and five cheap forwards run on it; this is the v1 two-stage I/O-bound argument, now with measured v2 throughputs.

The sky sweep: 17.3 M $\rightarrow$ 29,892 $\rightarrow$ FDR-controlled candidates. We swept 17,290,814 DR9 parent-galaxy positions (5,018,411 north, 12,272,362 south; 41 non-finite dropped) through the v2-lean stage-1 union filter at a 10^{-4} per-member FPR (`sweep/stage1_summary.json`). 29,892 survivors passed (1.73×10^{-3} realized rate, consistent with the EVT prediction within $\times 3$ of the five-member union bound), of which 1,440 crossmatch to known lenses and 28,452 are new. Group-conformal Benjamini-Hochberg selection (`sweep/conformal_summary.json`, `sweep/sweep_summary.json`) then certifies candidates: at per-group FDR ≤ 0.05 it selects 1,449 (813 new, 737 new-and-unseen after excluding training/mining/calibration rows), 2,836 at ≤ 0.10 , and 5,141 at ≤ 0.25 .

Conformal validity and the two-stage-selection trap. The selection’s validity rests on one subtlety we state plainly. The PRIMARY estimator computes each sweep row’s conformal p -value against the NegEval calibration with the *full* per-footprint sweep total as the multiple-testing count m (censoring non-survivors at $p=1$); because censoring can only raise a p -value and the unconditional p -value of a null sweep row is exchangeable with the calibration, the per-group FDR $\leq \alpha$ guarantee holds marginally. The tempting shortcut — using $m = n_{\text{survivors}}$ — conditions the test set on a stage-1 selection computed from the *same* members, is anti-conservative, and a `--synthetic-check` demonstrates the violation; it would have “selected” all 29,892 survivors at every α (the diagnostic row of Table 5). The honest cost of validity is power: with a conformal p -floor of $1/(n_{\text{cal},g}+1)$, the north footprint hits the power floor at $\alpha=0.05$ and selects *nothing* (surfaced, not hidden) — all 1,449 FDR ≤ 0.05 candidates are southern, where the calibration set is larger. NegEval is used twice (stage-1 thresholds and conformal calibration); a deterministic seed-2026 split confines calibration to a disjoint half and drops the 735 calibration rows that re-appear as sweep tests.

Known-lens recall consistency and “candidates, not confirmations.” The pipeline re-finds real lenses at the same operating point it proposes new ones: within the swept coverage it recovers 47.5% of in-coverage catalogued lenses into the survivor set (1,676/3,528 union; 40.3% Storfer, 59.0% Inchausti, 55.7% Huang21; `sweep/crossmatch_recall.json`), and 285 of them survive into the FDR ≤ 0.05 set — the internal consistency the new-candidate claim requires. We frame the output honestly (`sweep/visual_grading.md`): a first-pass visual grading of the top ~ 300 new candidates

finds them overwhelmingly luminous red galaxies with blends/companions and only $\sim 5\text{--}8\%$ showing plausible lens morphology (a tangential feature, partial ring, or arclet pair), because galaxy-galaxy $\theta_E \sim 1\text{--}2''$ is 4–8 px at DECaLS $0.262''/\text{px}$, at the resolution floor. *This is a statistically-controlled candidate list, not a confirmed-lens catalogue*: any individual candidate needs higher-resolution imaging (HSC/Euclid/HST) or spectroscopy to confirm. The deliverable’s value is the control — 737 new-and-unseen candidates at certified per-group FDR ≤ 0.05 from a 17.3M sweep, with a passing known-lens recall sanity check.

Equivariance, trained not test-time. v1 only tested test-time D_4 pooling (`equivariance.json`: +0.032 shielded, +0.049 Lanusse-46 Storfer@1% FPR). v2 replaces this with a genuinely *trained* equivariant member via `escnn` (Weiler and Cesa, 2019): a cyclic- C_4 pilot, gated against a self-calibrating parameter-matched plain-CNN twin (`141_pilot_escnn_c4.py`), beat the twin by +0.120 Storfer@1% FPR (`escnn_pilot.json`), clearing the +0.02 ship margin, which justified queueing a full dihedral- D_4 member (`142_train_escnn_d4.py`) as a bonus add to the roster.

Honest limits (v2). The v2 program tightens, but does not erase, the v1 caveats. “Recovery” is still the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly confirmed lenses; the 1,449/737 sky-sweep candidates are exactly that — candidates — and none is confirmed here. The matched-FPR deltas are now CI-backed down to 0.01% FPR (the headline’s largest gains), but the 0.01% threshold is pinned by ~ 100 NegEval negatives, so a residual PU-contamination assumption remains (bounded, not eliminated, by the top-200 audit). The conformal guarantee is marginal and per-group, assumes calibration/sweep exchangeability, and buys validity at the cost of power — the north footprint selects nothing at $\alpha=0.05$. The sweep’s stage-1 rate matches EVT only “within $\times 3$,” and the end-to-end wall-clock remains a structural I/O-bound argument rather than a benchmarked production run. The defensible v2 claim is sharper than v1’s: at the strict operating points scans actually use, v2-lean recovers +0.14 (Storfer@0.1% FPR) to +0.34 (Storfer@0.01% FPR) more catalogued lenses than the v1 flagship and +0.22 to +0.26 more than the published meta-learner, with a certified-FDR candidate list as the operational artifact — a measured, honestly-bounded, deployment-validated lift, not a transformational one.

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